

M.S., Student at KAIST AI

Contact Information

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Interest

My research focuses on making language models reason more reliably. I study the learning signals, data structures, and evaluation methods that help models generalize beyond superficial patterns, especially in compositional and multi-step reasoning. The goal is to build models whose reasoning is robust, transferable, and understandable without naively relying on data scaling.

Experience

Research Intern

AI Companion Team, KRAFTON AI, Seoul, Republic of Korea May 2025 – Nov 2025
 - Dept. Lead: Kangwook Lee
 - Topic: Developing Game-Play Agent (Co-playable Character) for PUBG (w. NVIDIA)
 (Link: <https://www.youtube.com/watch?v=wEKUSMqrbzQ>)

Research Intern

LK Lab, KAIST, Remote, Republic of Korea *Dec 2022 – Feb 2024*
 - Mentor: Seonghyeon Ye
 - Topic: Prompting Mechanisms of LLMs, Fine-grained Evaluation Metric

Research Intern

Naver CLOVA, Healthcare AI, Seongnam, Republic of Korea *Mar 2022 – Aug 2022*
 - Mentor: Yera Choi
 - Topic: EHR Summarization and embedding

Research Intern

KCVL Lab, KAIST, Daejeon, Republic of Korea
 - Mentor: Hangil Park
 - Topic: 2D Classification and Detection on imbalanced dataset

Education

KAIST

Ph.D., Graduate School of AI
Advisor : Seongjoon Oh

KAIST

M.S., Graduate School of AI
Advisor : Minjoon Seo

KAIST

B.S., School of Computing
Advisor : Tae-Kyun Kim
GPA: 3.96 / 4.3

Languages

- Korean	Native
- English	Fluent (TOEFL 113)
- Spanish	Fluent (DELE C1)

Publications

- [1] **Hwang, H.**, & Park, Y. (2026). Intrinsic Task Symmetry Drives Generalization in Algorithmic Tasks (**ICML 2026**)
- [2] **Hwang, H.**, Jeon, B., Kim, S., Kim, J., Chang, H., Yang, S., Won, S., Lee, D., Ahn, Y., ... (2025). *Let's Predict Sentence by Sentence*. arXiv preprint arXiv:2505.22202. (COLM RAM 2 Workshop [**Oral**])
- [3] Won, Y., Lee, H., **Hwang, H.**, & Seo, M. (2025). *Differential Information: An Information-Theoretic Perspective on Preference Optimization*. arXiv preprint arXiv:2505.23761. (Under Review)
- [4] Chang, H., Park, J., Cho, H., Yang, S., Ko, M., **Hwang, H.**, Won, S., Lee, D., Ahn, Y., ... (2025). *The Coverage Principle: A Framework for Understanding Compositional Generalization*. arXiv preprint arXiv:2505.20278. (ICLR 2026)
- [5] Lee, S., Kim, S., Seo, M., Jo, Y., Go, D., **Hwang, H.**, Park, J., Yue, X., Welleck, S., ... (2025). *The CoT Encyclopedia: Analyzing, Predicting, and Controlling how a Reasoning Model will Think*. arXiv preprint arXiv:2505.10185. (ICLR 2026)
- [6] Kim, J., Lee, H., Cho, H., Jang, J., **Hwang, H.**, Won, S., Ahn, Y., Lee, D., & Seo, M. (2024). *Knowledge Entropy Decay during Language Model Pretraining Hinders New Knowledge Acquisition*. (ICLR 2025 [**Oral**] & AAAI Workshop KnowFM 2025 [**Best Paper**])
- [6] Kim, S., Suk, J., Cho, J. Y., Longpre, S., Kim, C., Yoon, D., Son, G., Cho, Y., & **Hwang, H.** (2024). *The BiGGen Bench: A Principled Benchmark for Fine-grained Evaluation of Language Models with Language Models* (NAACL 2025 **BEST PAPER**)
- [7] **Hwang, H.**, Kim, D., Kim, S., Ye, S., & Seo, M. (2024). *Self-Explore to Avoid the Pit: Improving the Reasoning Capabilities of Language Models with Fine-grained Rewards*. (EMNLP Findings (2024) & ACL Workshop NLRSE 2024 [**Oral**])
- [8] Ye, S., **Hwang, H.**, Yang, S., Yun, H., Kim, Y., & Seo, M. (2023). *Investigating the Effectiveness of Task-Agnostic Prefix Prompt for Instruction Following*. (AAAI 2024)
- [9] Ye, S., Kim, D., Kim, S., **Hwang, H.**, Kim, S., Jo, Y., Thorne, J., Kim, J., & Seo, M. (2023). *Flask: Fine-grained language model evaluation based on alignment skill sets*. (ICLR 2024 [**Spotlight**])
- [10] **Hwang, H.**, Kim, S., Park, W. J., Seo, J., Ko, K., & Yeo, H. (2022). *Vision transformer equipped with neural resizer on facial expression recognition task*. (ICASSP 2022)

Honors & Awards

- Young-Han Kim Global Leadership Scholarship Awardee (2022)
- Qualcomm-KAIST Innovation Award Hackathon Winner (2023)